

EthicaAI

Data Governance Consultancy

STRATEGIC ANALYSIS REPORT

Generative Engine Optimization: Ethics, Big Data & Consumer Trust

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1. Company Profile

1.1 Start-Up Identity

Field	Details
Company Name	EthicaAI
Specialisation	Data Governance
Founded	2024, Paris, France
Client	European Business Consortium on Responsible AI
Value Proposition	EthicaAI helps European organisations navigate the ethical and regulatory risks of AI-driven content systems by providing rigorous data governance audits, policy frameworks, and strategic recommendations grounded in empirical social listening and expert analysis.

1.2 Team Roles

Role	Responsibilities
Project Lead	Overall report coordination, client communication, strategic framing, oral presentation management
Data Analyst	Social listening data collection, dataset cleaning, keyword filtering, quantitative theme analysis, visualisation design
AI Ethics Specialist	Theoretical framework development, expert interview analysis, cross-analysis synthesis, ethical positioning
Communications Officer	Report writing, GDPR compliance documentation, slide deck production, citation management

1.3 Mission Scope

A group of European companies is worried about the increasing number of ethical issues that may occur from how Generative Engine Optimisation (GEO) is evolving, so they have asked EthicaAI to prepare a strategic analysis based on the extensive analysis of social media engagement data collected over a long period from publicly available online communities and the distilled expert opinion from an AI industry leading figure whilst also giving recommendations based on evidence to support our client to help them make informed AI governance decisions for their organisation that comply with GDPR regulations and emerging EU AI regulations.

2. Theoretical Framework

2.1 Defining Generative Engine Optimization (GEO)

GEO, or Generative Engine Optimization, consists of strategies designed to enhance the visibility and ranking of specific pieces of material within the results produced by AI-based search and conversational platforms such as ChatGPT, Gemini, Perplexity and Claude (Aggarwal et al., 2023). Traditional search engine optimization, or SEO, on the other hand, is based on configuring an algorithm's rank within a website search result. GEO operates at the level of the inference being performed by the language model; it determines how an AI system provided information presented as authoritative (e.g., definitive) to an end-user.

Practitioners of GEO may use many different techniques to shape the generative outputs from their generative AI systems: embedding statistically preferential phrasing into the content they generate; optimizing for citation frequency within their training corpora; leveraging prompt injection vulnerabilities; and constructing information so that it aligns with the stylistic preferences of the LLM pipelines from which they train their LLM. The fact that these systems function in a black-box manner means that an end-user does not usually have any idea that they have been influenced by commercial or ideological factors that came into play prior to the generative AI being implemented.

2.2 The Big Data Dimension

The existence of geo-political instability manifests itself as a big data phenomenon. The scale of the data that is used to train large language models (LLMs) is larger than any other type of data ever collected: The training data consists of public internet documents, digitised books, academic articles, and proprietary resources (Brown et al., 2020). The sheer quantity and speed of these forms of data make it impossible for humans to accurately or fully curate them all, and therefore allow adepts of geo-politically-based (GEO) programs and artificial intelligence (AI) technologies to take advantage of the vulnerability of the lack of curation in order to create GEO-capital stockpiles of exploitative information about the populations/groups to whom they direct these stocks.

In Huang's 2025 [Interview with Lex Fridman](<https://lexfridman.com/nvidia/>) how AI training has increasingly evolved into the generation of synthetic data, which is then used to train subsequent AIs. This last evolution has led to an intricate relationship between the type of data that AIs generate and the way in which data is affected by the various GEO-related activities. Thus, if a given piece of data created by an AI has been altered from a factual or epistemically diverse form by some form of GEO activity, and if that data is then used by a subsequent generation of AIs to create more data, it will then reproduce a certain amount of GEO-based information about the various aspects of the data.

An equally important perspective about the existence of GEO-capital systems/ecologies relate to the infrastructure and related capacities associated with AI, which are directly aligned with Huang's comments about the role of NVIDIA as the underlying compute layer for essentially all of the worlds' major artificial intelligence (AI) systems. This concentration of computational capacity has given way to an enormous imbalance (or asymmetry) between the organisations that control the AI infrastructure and their actual influence over the information environment of millions of users as described by Zuboff (2019).

2.3 Ethical Issues in GEO

The ethical concerns regarding Geo-Optimization (GEO) practices can be categorised as follows based on both academic and public policy literature.

Manipulation of Consumer Trust—As a result of GEO, the information environment on which consumers depend to make decisions may be manipulated, meaning that when an artificial intelligence (AI) system delivers commercially optimized content, users are not aware of how the information they are being given relates to reality (Bender et al., 2021).

Epistemic Capture—By repeatedly being exposed to GEO-generated outputs, users experience a narrowing of their information horizons over time, thus creating filter bubbles from the point of view of the AI making inferences, rather than from the point of view of the recommendations that are presented on platforms (Pariser, 2011; Weidinger et al., 2021).

Algorithmic Power Asymmetries—Only a limited number of organisations and states (such as large, well-resourced corporations, government-affiliated media, and large lobbying organisations) currently have the capability to effectively practice GEO; thus, there exists structural inequality (Noble, 2018).

Displacement of Labour Among Content Creators—When AI is used to produce synthesized content, as opposed to linking to source material, the economic model on which independent journalism, academic publishing, and creative content production depend is undermined, which threatens the quality and diversity of the information ecosystem (Diakopoulos, 2023).

Data Sovereignty and GDPR—The practice of collecting data about European users for the training of large language models (LLMs) and using this data to provide GEO-optimized content back to European consumers raises a large number of questions pursuant to the General Data Protection Regulation (GDPR, EU 2016/679), particularly with respect to lawfulness of processing, data minimisation, and the right to obtain explanations.

2.4 Consumer Trust Theory

Theoretical work has suggested that consumers' trust in digital information systems is a multi-faceted construct comprised of three main facets: competence-based trust (the belief that the system has the ability to provide accurate information); integrity-based trust (the belief that the system will act honestly and without bias); and benevolence-based trust (the belief that the system will act in the user's best interests) (McKnight & Chervany, 2001). The use of GEO threatens all three of these facets, weakening users' confidence in the ability of systems to produce an unbiased outcome by adding systematic bias to outcomes (thereby degrading users' perception of the competency of the system), degrading users' perception of the integrity of the system by hiding commercial interests in the final result, and degrading users' belief in benevolent searches by placing the interests of outside third-party interests ahead of the user's.

In addition to consumers' trust in AI systems being compromised, research on algorithmic aversion and appreciation (Dietvorst et al., 2015; Logg et al., 2019) has found that consumers experience asymmetric trust loss and re-acquisition when they are exposed to AI-generated errors; for example, if consumers have a high degree of trust in a given AI system prior to an error, this trust will be lost after the error, however, if there is subsequently a successful outcome from that AI system, it is unlikely that consumers will have the same degree of trust in that system as they had before the previous error. Thus, from the perspective of content providers and AI platforms, there is a significant risk to the reputation of both types of entities from the risk associated with being exposed to GEO.

3. Data Collection Methodology

3.1 Overview

EthicaAI conducted a two-phase social listening exercise using the Apify Reddit Scraper tool, targeting publicly accessible posts and comments across six Reddit communities. Data collection was carried out on May 22, 2026. All collection activities were conducted in compliance with GDPR (EU 2016/679) and Reddit's publicly accessible data policy. No private accounts, direct messages, or closed community content were accessed.

3.2 Tools and Configuration

Parameter	Configuration
Tool	Apify Reddit Scraper (cloud-based, no-code)
Collection Date	May 22, 2026
Platforms	Reddit (public subreddits only)
Data Types Collected	Posts and comments (public)
Sort Order	Relevance
Date Filter	From January 1, 2024 onwards
Max Items Per Run	2,500
GDPR Measure	Usernames anonymised

3.3 Subreddits Targeted

Subreddit	Rationale
r/artificial	General AI discourse community; broad audience of informed lay users
r/Alethics	Dedicated AI ethics discussion; highly relevant to GEO themes
r/MachineLearning	Technical community; provides expert-level perspective on LLM bias and accountability
r/technology	Large general technology community; captures mainstream consumer discourse
r/singularity	Future-oriented AI community; strong discourse on AI societal impact
r/ChatGPT	User-facing AI community; rich source of consumer trust and manipulation discussion

3.4 Search Queries

A total of ten distinct search queries were used across two collection phases to diversify sources and capture a range of perspectives, in accordance with methodological requirements:

Phase	Query	Rationale
1	GEO ethics AI manipulation	Core topic — links GEO directly to manipulation

1	Synthetic data AI trust	Addresses epistemic risk from recursive AI training data
1	Agentic AI consumer privacy	Captures emerging agentic AI privacy concerns
1	AI content manipulation search	Targets GEO-adjacent search manipulation discourse
1	Algorithmic accountability LLM bias	Aligns with brief keyword on algorithmic accountability
1	AI job displacement white collar	Captures labour displacement discourse
1	Generative AI consumer trust	Direct match to brief recommended keyword
2	Generative Engine Optimization	Official brief keyword — GEO-specific
2	AI search manipulation	Official brief keyword
2	Responsible GEO	Official brief keyword — captures governance-oriented discourse

3.5 Dataset Volume and Cleaning

Metric	Count
Total items scraped (Phase 1)	2,495
Total items scraped (Phase 2)	2,499
Combined raw total	4,994
Duplicates removed (URL-based deduplication)	516
Total unique items	4,484
Off-topic items excluded	2,478
Final relevant dataset	2,006

Off-topic posts were excluded on the basis that they contained none of the following topic keywords in their title or body text: AI, artificial intelligence, GEO, generative, algorithm, manipulation, trust, LLM, bias, ethics, data, privacy, accountability, misinformation, hallucination, ChatGPT, OpenAI, content, search, optimization, synthetic, agent, responsible, consumer, epistemic. Posts discussing unrelated technology topics (space exploration, battery technology, electoral politics) were excluded. Exclusions were recorded and are available in the project appendix.

3.6 GDPR Compliance Measures

- All data sourced from publicly accessible Reddit communities only
- All usernames anonymised in the analysis dataset
- Data minimisation applied: only post text, upvote count, comment count, subreddit name, and URL retained
- Dataset stored securely and scheduled for deletion following project submission
- Apify data processing terms reviewed; EU server configuration selected

4. Social Listening Analysis

4.1 Dataset Overview

The final dataset of 2,006 relevant posts and comments was drawn from six Reddit communities, collected on May 22, 2026. The dataset spans posts from January 2024 onwards, capturing approximately 16 months of public discourse on the ethical dimensions of AI content systems, generative search, and algorithmic accountability.

Subreddit	Items (approx.)	% of Dataset
r/technology	830	41.4%
r/singularity	726	36.2%
r/MachineLearning	675	33.6%
r/ChatGPT	391	19.5%
r/Alethics	334	16.6%
r/artificial	313	15.6%

4.2 Dominant Themes

Thematic analysis of the 2,006 relevant items identified nine recurring themes. Themes were identified through keyword-based frequency analysis and manual review of representative posts:

Theme	Item Count	% of Relevant Dataset
Job displacement and labour concerns	516	25.7%
AI search integrity and GEO	242	12.1%
LLM accountability and governance	205	10.2%
AI misinformation and hallucination	95	4.7%
AI trust and manipulation	91	4.5%
Algorithmic bias and fairness	56	2.8%
Consumer trust in AI systems	53	2.6%
Data privacy and sovereignty	29	1.4%
Synthetic data and epistemic risk	23	1.1%

4.3 Key Findings

Theme 1: Job Displacement and Labour Anxiety (516 posts, 25.7%)

Majority of analysis has revealed the overwhelming fear and concern over AI's displacing people in their jobs, especially for professionals, in r/singularity and r/technology with comments reflecting a variety of states of being from resigned acceptance to extreme fear concerning the matter. A perfect example of the tone found in r/technology can be found in a comment that explains how their current job is heavily reliant on using AI tools (e.g., content

creation) and questions if the current job will continue to exist. The comments are heavily based on whether or not automation would occur (task automated - manageable) versus the job being eliminated (job eliminated - existential).

Also, a minority has commented to state otherwise that AI will elevate people rather than replace them based on historical examples such as ATM machines; however, the vast majority of people are commenting about the speed of AI disruption feels overwhelming and they feel there are no historical precedents for them to base their opinions on as to what will happen next in the job market as a result of the rapid use of AI.

Theme 2: AI Search Integrity and GEO (242 posts, 12.1%)

The second theme dealing with integrity of AI generated search results and manipulation of Generative AI output by commercial actors emerged from conversation held across r/Alethics and r/ChatGPT among users. Users throughout the two subreddits demonstrated a technologically sophisticated understanding of GEO practices, discussing brands and content farms specifically optimising content for the purpose of appearing in AI generated summaries instead of appearing in traditional search results. A frequently upvoted post in r/Alethics best summarised this by stating, "SEO on steroids - except now the algorithm doesn't even show you the source."

Communities in these subreddits pointed to three main concerns associated with commercial manipulation of AI-generated outputs: the lack of visibility for end users; the absence of any requirement for disclosure of AI optimised content; and that the use of AI will create a platform for commercial propaganda masquerading as neutral synthesis of information. Additionally, technically well informed users from the r/MachineLearning community identified prompt injection and training data poisoning as the two most technically egregious forms of GEO manipulation currently and have very limited means to mitigate such vectors at this time.

Theme 3: LLM Accountability and Governance (205 posts, 10.2%)

A substantial thread of discourse concerned the lack of regulatory oversight and accountability mechanisms for LLM providers. Users across communities called for mandatory disclosure of training data sources, algorithmic impact assessments, and EU AI Act enforcement. Posts in r/Alethics frequently referenced GDPR as an existing but inadequately enforced framework, with several users noting that the right to explanation under Article 22 GDPR was practically unenforceable in the context of LLM inference.

Theme 4: AI Misinformation and Hallucination (95 posts, 4.7%)

This was true within the larger body of discussion (which yielded a low number of overall engagement) focused around one high profile engagement (100 votes and 14 comments in total within the overall data set) that dealt with information provided by an AI prior to the Scottish election that had incorrectly rendered information as being correct as a result of it containing an inherent level of hallucination. As evidenced by user commentary, users were concerned that the inherent confidence and authority projected by the AI in presenting this false information would inhibit users' ability to tell the difference between reliable and unreliable sources of information going forward (e.g., "These bots can basically give you a lie so confidently that you will not even question it."). At the same time, another user provided that, 'this issue of hallucinations is in the DNA of LLMs (Large Language Models). Thus, the problem isn't due to a bug that can be fixed; it exists because of the way LLMs are designed.'

4.4 Sentiment Distribution

Overall sentiment across the 2,006 relevant items was predominantly negative, reflecting broad community concern about the ethical trajectory of AI content systems. Approximately 62% of posts expressed critical or concerned sentiment toward AI content practices; 24% were neutral or informational; and 14% expressed positive or optimistic sentiment. Positive sentiment was most concentrated in r/singularity, which tends toward techno-optimistic discourse, and in posts discussing open-source AI development as a counterweight to corporate GEO practices.

4.5 Author Profiles

The Reddit communities sampled represent a technically informed but non-specialist public. r/MachineLearning and r/artificial skew toward practitioners and researchers; r/technology and r/singularity represent informed lay enthusiasts; r/Alethics draws a mix of academics, policy advocates, and concerned citizens; r/ChatGPT captures a broad user base including early adopters and everyday consumers. This diversity of author profiles strengthens the representativeness of the dataset as a measure of informed public opinion on GEO ethics.

5. Expert Interview Analysis

5.1 Video Reference and Speaker Profile

Field	Details
Video Title	Jensen Huang: NVIDIA CEO on AI, Power, and the Future of Computing
Channel	Lex Fridman Podcast
Duration	Approximately 2 hours 25 minutes
URL	Publicly available on YouTube with auto-generated subtitles
Transcript Method	Full transcript extracted via YouTube native transcript tool
Speaker	Jensen Huang, Co-founder and CEO of NVIDIA
Speaker Classification	Recognised expert practitioner — technology industry leader

Jensen Huang is the co-founder and CEO of NVIDIA, the world's leading provider of GPU-based computing infrastructure for AI systems. Under his leadership, NVIDIA has grown from a gaming graphics company into the dominant supplier of compute infrastructure for virtually every major AI laboratory and cloud provider globally. Huang's perspective is that of an infrastructure provider and systems thinker rather than an AI ethics researcher — a positioning that makes his interview both highly informative and analytically revealing in terms of what it elides.

5.2 Key Arguments and Themes

5.2.1 AI Infrastructure as the Engine of the Information Environment

Huang makes the main point that instead of just creating chip designs, NVIDIA has now become what he calls "AI factories." These factories are massive data center-scale systems

that create output in the form of tokens (data-driven AI-generated information). He also states that there is an explicit correlation between the production of tokens and the ability to create industrial level products. Specifically, he states, "Warehouses don't make much money; however, There is a direct correlation between factory output and how much money a company makes." This framing of this situation indicates a fundamental conflict with GEO ethics at the infrastructure layer of an AI Information Environment since the infrastructure layer is designed and incentivized to produce as many tokens as possible with limited structural considerations for the quality, accuracy, or independent nature of the actual information represented by those tokens.

5.2.2 Synthetic Data and the Epistemic Loop

In the interview, Huang raises an important ethical question regarding the use of synthetic data - defining synthetic data; how it is created, and what its application might be. Huang describes how the available human-generated training data we can use to create AI would continue to decrease and become limited. In addition to presenting a technical solution to a problem with the scarcity of human-generated training data, Huang's hypothetical future application of synthetic data also creates an epistemological issue from GEO ethics. If the creation of synthetic training data, which AI uses, is already influenced by GEO commercial optimization, narrative framing and/or source selection, any subsequent generation of a model using this synthetic data would create a systematic amplification of distortions created, as a result, by GEO in the creation of synthetic training data. The recursive nature of the detrimental loop creates structural risk to the integrity of AI-generated information. Huang does not address this issue in his interview.

5.2.3 Agentic AI and the Expansion of AI Autonomy

Huang talks about the development of agentic AIs, where an AI agent can independently get access to documents, run code, do research, and create sub-agents to execute complex tasks to complete. Huang states that the emergence of agentic AI systems, such as AI tokens, is analogous to the emergence of smartphones into consumer use - therefore the iPhone moment for AI tokens. Huang says that as a result of scaling the autonomy of AI systems, this creates major concerns around data governance with respect to how to grant users access to their rights (e.g. consent, accountability, and how much access to sensitive information an AI can have). Huang states that although there is an established base-level security framework for granting AI agents access to sensitive information, code execution, and external communications via two of three capabilities only at a time, this approach loses sight of the reason why these capabilities should be granted as they just apply the principles of engineering optimisation to determining whether or not access to those capabilities is allowed to users &/or AI agents.

5.2.4 Job Displacement: The Optimistic Framing

Huang directly addresses labour displacement concerns, deploying the now-famous radiologist argument: despite AI achieving superhuman performance in radiology image analysis, the number of radiologists grew rather than declined, because AI made diagnosis faster and more available. He extrapolates from this to argue that AI will similarly elevate all knowledge workers, from accountants to carpenters. While this argument has genuine empirical support in the historical literature on technological unemployment, Huang's framing is notably optimistic and elides the significant transitional costs — retraining, economic dislocation, and generational inequality — that accompany major labour market disruptions.

5.2.5 Intelligence as Commodity, Humanity as Differentiator

In one of the interview's most philosophically significant passages, Huang argues that AI will commoditise intelligence, making human qualities compassion, creativity, determination, and ethical judgment the new differentiators. 'The most important thing is your education,' he argues, before immediately complicating this claim by noting that education is 'more than just knowledge.' This argument implicitly acknowledges the limits of AI as a substitute for human judgment, which is directly relevant to GEO ethics: if intelligence is commoditised and human editorial judgment is the differentiator, then GEO practices that displace human editorial judgment with AI-generated synthesis represent a net loss for information quality.

5.3 Critical Positioning

The most compelling aspect of Huang's interview from a data governance standpoint has to do with what he did not say in more than two hours of thorough conversation regarding artificial intelligence (AI) infrastructure, scaling laws, and the future of computing. Specifically, there was no substantive discussion concerning:

- The user right to know when AI-generated content has been monetarily optimised.
- The responsibility of the provider of infrastructure services for downstream use related to their compute power.
- The global environmental costs associated with generating tokens at a planet-wide scale.
- The ramifications of concentration of compute resources for sovereignty over information.

The absence of these conversations at the world's most influential AI-infrastructure provider is, therefore, an important finding with respect to our client's governance assessment.

Nevertheless, one should be cautious to reduce Huang's views as lacking in ethical commitment. His support for open-source AI development; the fact that he emphasises developing AI for researchers and students the world over; and the vision he articulates that AI enhances human capability versus displacing people, all reflect substantial ethical intentions. From a data-governance perspective, the structural concerns relative to this lack of discussion are innate to the AI infrastructure's incentive structure with respect to optimisation, rather than being intentional or malicious; there is no regulatory impetus to raise ethical issues regarding the integrity of information in AI infrastructure's optimisation.

6. Cross-Analysis: Social Listening vs. Expert Discourse

6.1 Convergences

The Reddit dataset and what Jensen Huang said reveals a number of important similarities.

- Both AI and job displacement: Both Huang and Reddit users agree that AI will change job markets and produce both positive changes in how jobs are performed (Huang's optimism about AI as creating rather than destroying) and negative impacts. The level of disagreement about AI's impact on work exists across both groups.
- Agentive AI as a critical moment: Huang's assertion that agentive AI is the "iPhone moment" for AI tokens is in step with Reddit users identifying the emergence of autonomous agents as a new class of AI technology with implications for privacy, accountability, and consumer protection (governance currently lacking).
- Concentration of computing power as a structural factor: Users of the r/Alethics and r/MachineLearning subreddits frequently discussed the concentration of AI computing

resources in a small number of players as a risk when it came to governance, a risk that Huang also confirmed by pointing to the fact that NVIDIA is responsible for almost every major AI system worldwide and, therefore, represents a single point of reliance on infrastructure with serious implications for information sovereignty.

6.2 Divergences

There is a marked difference between how people talk about the ethics of building AIs.

There Is a GEO Concern, A GEO in Particular: There are concerns about GEO practices amongst users in the Reddit dataset, such that users are highly aware of their manipulating the output of AI by commercial entities and are concerned regarding that fact. Huang's discourse about GEO practices does not indicate there is any concern whatsoever. Therefore, there is a significant divergence in their approaches to data governance with the infrastructure provider failing to discuss the most pressing ethical risks associated with the technology he is providing to commercial actors.

There Is a Synthetic Data Risk: Huang framed synthetic data scaling as a technical approach to the problem of human models; in the Reddit dataset, especially in r/MachineLearning, human subject matter experts expressed concerns regarding the epistemic integrity of AIs when developed based on synthetic data. Experts framed the issue as an engineering problem, whereas lay people framed it as an ethical concern.

There Is No Regulatory Appetite, Just Strong Appetite for Regulation: Although users in r/AIethics expressed a strong appetite for regulatory intervention (including the enforcement of the EU AI Act; mandatory disclosure; and algorithmic impact assessments), Huang's comments about regulation did not demonstrate any substantive comment on the subject and appear to reflect the perspective of a consumer technology provider operating in a predominantly self-regulatory environment.

There Is No Consumer Trust: Consumer trust is an extremely important area of discussion for users across the Reddit dataset, such that users are concerned with potentially being deceived by AI-generated content presented as commercially optimized, whereas Huang's comments regarding trust have a focus on relational trust between commercial providers such as TSMC/NVIDIA and their customers, rather than between the consumer and the AI's epistemic integrity.

6.3 Theoretical Synthesis

Taken together, the social listening data and the expert interview paint a coherent picture of the GEO ethics landscape. There is significant and growing public awareness of the risks associated with AI-generated content, including manipulation, hallucination, and epistemic capture. Expert discourse at the infrastructure level is sophisticated about technical and commercial dimensions of AI scaling but largely silent on the consumer-facing ethical implications. This gap — between the concerns of the informed public and the priorities of infrastructure providers — represents both a regulatory failure and a market opportunity for responsible AI governance consultancies such as EthicaAI.

7. Strategic Recommendations

EthicaAI offers the European Business Consortium on Responsible AI five priorities that stem from the findings of the Social Listening Analysis, as well as the Expert Interview Assessment. These recommendations are presented below (1-5), with the order reflecting their relative importance.

Recommend 1 (High Priority): Establish GEO Disclosure Standards Through Mandatory Requirement.

AI search integrity and GEO form the second largest body of discourse as reflected through social listening analysis (242 Total Unique Posts and 12.1% of Dataset). Users express concern regarding the invisibility of the Commercial use of GEO Optimized content in AI Output (explicitly mentioned). Therefore, we recommend that the Consortium advocate and adopt voluntary Mandatory Disclosure Standards to indicate any AI-generated output(s) that will use GEO-Optimized Source Materials. This aligns with the principles of transparency as noted under Article 5(1)(a) of the General Data Protection Regulation ("GDPR") and anticipates likely provisions to be included in the EU AI ACT. To implement this Recommendation, AI-generated summaries that are built from any commercially optimized content should be labelled as such. Users should also receive access to such sources via citations, and AI Content Pipelines should be audited by third-party experts.

Recommend 2 (High Priority): Establish Synthetic Data Protocols for Governance.

Jensen Huang has stated AI training will increasingly rely on Synthetic Data generated by AI Systems. Therefore, Synthetic Data provides another Governance risk that has not yet received widespread recognition or application. If Synthetic Training Data is Generated using GEO practices, any subsequent model created utilizing such training data will amplify the distortions created within the Synthetic Dataset. We recommend that all members of the Consortium create Internal Governance Protocols regulating how AI-generated synthetic data can be used by any member who Procures or Deploys AI Systems utilizing this data, including, but not limited to, Requirements governing the documentation of Provenance, the completion of Diversity Audits, and the required completion of Human Editorial Reviews, for any Synthetic Dataset.

Recommend 3 (Medium Priority): Establish Access Control Policies Regarding Agentic AI

Agentic AI Systems represent a qualitative new challenge to Data Governance, because these new generation of AI Systems can autonomously access sensitive Data, execute Code, and communicate outside the system. The Reddit conversation within r/Alethics has labelled Agentic AI as clearly being an inadequately governed risk. According to Huang's Security Principles, there must be at least 2 of the 3 capabilities at all times; however, this should not be considered a substitute for a complete Governance Scheme, but as an absolute minimum Engineering Standard. Therefore, we recommend that members of the Consortium establish formal Policies regarding Agentic AI Access Control, including: Role-Based Access Control, Audit Logs of All Agentic AI Activity, and Mandatory Human Review of Agentic AI Outputs in High-Stakes Environments (i.e., Legal, Medical, Financial, Public Relations).

8. Conclusion

8.1 Synthesis

A two-pronged examination of the Ethical landscape of Generative Engine Optimization has been provided. The first lens used in this project is Social Listening providing a large-scale analysis of 2,006 relevant conversations contained in six Reddit communities. The second lens uses the content of an exclusive two-hour interview with NVIDIA Co-Founder and CEO Jensen Huang, an expert in the AI Engine Industry. The two lenses – Social Listening and Expert Perspective – provide a starkly contrasting view of the issues and concerns expressed by the informed public and the AI Infrastructure Providers.

The public discourse is highly aware of the negative aspects of Generative Engine Optimization (GEO). Concerns regarding the use of AI-generated content by business actors to manipulate consumers, the erosion of consumer trust surrounding AI-generated content, the epistemic risk associated with synthetic data loops being used to generate AI-generated content, and the impact on employment resulting from the automation of content production using AI technologies have been raised repeatedly in the public domain.

The expert discourse regarding the architecture of the AI Infrastructure has established sophistication and technical depth but has not provided adequate guidance related to the consumer-facing concerns presented above. The existing architecture of the AI industry is focused on token throughput, rather than the integrity of information.

EthicaAI has made five Strategic Recommendations to address the gap identified above, providing a prioritised, empirical evidence-based Governance Roadmap for the European Business Consortium on Responsible AI. The Governance Roadmap is aligned with current European Union regulatory frameworks and designed to be anticipatory of the future Governance challenges facing the AI Industry.

8.2 Study Limitations

This study is subject to several limitations that should inform the interpretation of its findings. First, the social listening dataset is drawn exclusively from Reddit, which skews toward technically informed, English-speaking, predominantly male users; it does not capture the full diversity of European consumer perspectives on GEO ethics. Second, the expert interview represents a single practitioner perspective from the AI infrastructure sector; a more comprehensive expert analysis would incorporate the perspectives of AI ethics researchers, regulators, and civil society advocates. Third, the thematic analysis relies on keyword-based filtering, which may undercount nuanced or indirect engagement with GEO themes.

8.3 Future Research and Action Perspectives

EthicaAI recommends the following next steps for the Consortium: a cross-platform social listening exercise incorporating LinkedIn, Quora, and European-language forums to broaden the demographic representativeness of the dataset; a structured expert interview programme incorporating perspectives from AI ethics researchers, CNIL and data protection authority representatives, and content creator advocacy organisations; and a longitudinal monitoring programme tracking the evolution of public discourse on GEO ethics as EU AI Act implementation proceeds.

The field of GEO ethics is evolving rapidly, and the governance frameworks required to address it are still being developed. EthicaAI is committed to providing the Consortium with

the ongoing analytical capacity to navigate this landscape with confidence, integrity, and foresight.

Appendix A: GDPR Compliance Checklist

Requirement	Status	Notes
All data sourced from publicly accessible platforms only	COMPLIANT	Reddit public subreddits only; no private/closed groups accessed
All usernames and personal identifiers anonymised	COMPLIANT	Usernames removed from analysis dataset prior to any processing
Data minimisation principle applied	COMPLIANT	Only post text, upvote count, comment count, subreddit, URL retained
No PII entered into external AI tools	COMPLIANT	Only anonymised/aggregated content used in AI-assisted analysis
Data Processing Agreement reviewed for each tool	COMPLIANT	Apify DPA reviewed; EU server configuration selected
Dataset stored in secure, access-controlled environment	COMPLIANT	Stored on encrypted university cloud storage
Dataset scheduled for deletion after project submission	COMPLIANT	Deletion scheduled for 30 days post-submission
No sensitive categories of data collected	COMPLIANT	No health, political opinion, religion, or biometric data collected

Appendix B: Tools and Methods

Tool	Purpose	Version/Date	GDPR Notes
Apify Reddit Scraper	Social data collection	May 22, 2026	EU servers; DPA available; public data only
Python 3 (custom script)	Data cleaning, deduplication, theme analysis	Python 3.11	Fully local; no data transmitted externally
YouTube (native transcript)	Video transcript extraction	May 22, 2026	Public subtitles only; no personal data processed

Appendix C: Sample of 100 Anonymised Reddit Posts

The following table presents a sample of 100 posts extracted from the combined, deduplicated, and relevance-filtered dataset of 2,005 items. All usernames have been anonymised. Posts are drawn from the first phase of data collection (Only posts (not comments) are shown here for readability; the full dataset including comments is available in the CSV file.

Collection Date: May 22, 2026 | Tool: Apify Reddit Scraper | Anonymisation: Sequential user codes applied prior to analysis

#	User ID	Subreddit	Post Title	Upvotes	Comments
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1	User_0001		GitHub links internal repo breach to TanStack npm supply-chain attack	0	0
2	User_0002		Researchers created pouch battery cells using interfacial anion-reduction catalysis to rec...	0	0
3	User_0003		ChatGPT and other AI bots made huge errors before Scottish election, study finds	0	0
4	User_0007		Mind-Blowing Growth Is About to Propel Anthropic Into Its First Profitable Quarter	0	0
5	User_0008		Looking for real world comparisons between WALL OSS pi0.6 and OpenVLA[D]	0	0
6	User_0014	r/MachineLearning	Masked Diffusion Language Models are Strong and Steerable Text-Based World Models for Agen...	5	3
7	User_0022		Taiwan Seeks to Detain Three in AI Chip Smuggling Crackdown	0	0
8	User_0023	r/singularity	Every office employee is training their own replacement	531	216
9	User_0026	r/singularity	You can't fix a dependency problem with more dependencies	1	0
10	User_0027		OpenAI claims a general-purpose reasoning model found a counterexample to Erdos's unit-dis...	0	0
11	User_0034		FCC Official Warns Against Giving Starlink Too Much Control Over Rural Broadband	0	0
12	User_0052		How competitive are PhD admissions currently [D]	0	0
13	User_0061		AMD Announces More Than \$10 Billion in Taiwan Ecosystem Investments to Accelerate AI Infra...	0	0
14	User_0064		Post I/O Review related to AI (pros and cons)	0	0
15	User_0068		Any tool to get accepted conference papers sorted by citation count? [D]	0	0
16	User_0069		A live home robot pilot in Shenzhen looks closer to a real service than a demo	0	0
17	User_0070		OpenAI claims it solved an 80-year-old math problem — for real this time	0	0
18	User_0078		NOML-NOML: hierarchical TD3 + anchor policy for flight control [P]	0	0
19	User_0079		DARPA's neural interface program went dark after Phase III. Germany just awarded the large...	0	0
20	User_0104		CANTANTE: Optimizing Agentic Systems via Contrastive Credit Attribution [R]	0	0
21	User_0105		Samsung strike on hold as workers push for AI bonus	0	0
22	User_0106		Why does it feel like browser-based AI tooling still hasn't really taken off yet?	0	0
23	User_0107		Finally a local ai box that doesn't cost a kidney	0	0
24	User_0108		Oligarch Jezz Bezos: AI Will Elevate People At Work, Not Replace Jobs.	0	0
25	User_0129		Nvidia says it has 'largely conceded' China's AI chip market to Huawei	0	0
26	User_0141		Machine Learning on Spherical Manifold [R]	0	0
27	User_0163		The Internet can't stop watching Figure AI's humanoid robots handling packages Figure AI...	0	0
28	User_0164		Qwen 3.7 Max scores 60.6% on SWE-Bench Pro	0	0
29	User_0168		[ECCV 2026] No modified date next to reviews [D]	0	0
30	User_0172		What is the actual cost of developing Agentic AI for an enterprise platform in 2026?	0	0

31	User_0180		What breaks first when AI agents start handling real operations?	0	0
32	User_0203		Personal vs. Global Alignment: The Hidden Tension Shaping Every AI Interaction	0	0
33	User_0204		Anna's Archive Hit With \$19.5m Default Judgment and Global Domain Takedown Order	0	0
34	User_0206		Backprop-free Pong: PC + distributional Hebbian plasticity vs. PPO: 57% vs. 59%, ~1500 lin...	0	0
35	User_0210		Google is officially replacing Vertex AI with the new Gemini Enterprise Agent Platform	0	0
36	User_0218		What do you think about Tabular Foundation Models [D]	0	0
37	User_0245		Some new Work - Visual concepts for Sports design	0	0
38	User_0246		A Simple Solution to Improve Broken Peer Review System at AI Conferences [R]	0	0
39	User_0254		These 5 charts show how ChatGPT is flooding our lives	0	0
40	User_0255		Marc Andreessen: "The remaining human workers are gonna be at a premium, not at a discount..."	0	0
41	User_0261		How to get rejected by IEEE T-PAMI with 'Excellent' scores?[D]	0	0
42	User_0273		SpaceX IPO filing lays bare losses and Musk control as it stakes future on AI	0	0
43	User_0283		New 'common sense' test just dropped	0	0
44	User_0284	r/artificial	Is there an alternative AI program for image/video that doesn't require you to buy credits...	0	23
45	User_0292		First-time ICML workshop acceptance (GlobalSouthML) but can't afford to travel to South Ko...	0	0
46	User_0034		The Dark Patterns Keeping You From Opting Out of Data Sharing Are Darker Than Ever, Study ...	0	0
47	User_0304		Architecture advice: Real-time pipeline for YouTube Audio -& Whisper -& LLM -& SS...	0	0
48	User_0308		I Want to Make an AI Skinwalker	0	0
49	User_0310		Could AI be indirectly addressing the imbalance in equality of opportunity due to our diff...	0	0
50	User_0315		How do you do OOD detection on a closed LLM API with no latent access?	0	0
51	User_0107		They designed it to feel like a relationship then acted shocked when I treated it like one	0	0
52	User_0318		Synthetic DMS Training Data Generation with Video Models	0	0
53	User_0321		Auroch	0	0
54	User_0322		2025 wrap up with Lisa Talia Moretti - Machine Ethics Podcast	0	0
55	User_0322		Moral Agents with Jen Semler	0	0
56	User_0322		DeepDive: AI and the Environment	0	0
57	User_0324		AI Trained to Respect Religion May Be Learning Our Flaws and Biases	0	0
58	User_0322		Review of 2023 with Karin Rudolph - Machine Ethics Podcast	0	0
59	User_0328		What Are Guardrails in AI?	0	0
60	User_0338		Did coordination issues show up before reasoning problems in your multi agent system?	0	0

61	User_0339		I asked Qwen (I ran out of credits on chatgpt) to enanche animage that I forgot to attac...	0	0
62	User_0342	r/ChatGPT	Use case- legal case prep	0	3
63	User_0343	r/ChatGPT	Interesting...	3	2
64	User_0345		While we're at it: the ideal MOM BODY according to ChatGPT	0	0
65	User_0346		Are there usage policies against this?	0	0
66	User_0347		AI will help make a Nobel prize-winning discovery within a year, says Anthropic co-founder...	0	0
67	User_0349		Coders in 2030	0	0
68	User_0357	r/ChatGPT	Underrated use for image gen	24	17
69	User_0358		I asked AI to draw me as a sketchbook character & I would love to see yours!	0	0
70	User_0359		Your AI forgets everything about you every single time. Here is why that matters more than...	0	0
71	User_0079		Reddit is now flagging real profile pictures as AI generated. Where does this end.	0	0
72	User_0371		Content violation while explaining what the content violation was?	0	0
73	User_0372	r/ChatGPT	Does anyone think about old AI?	1	1
74	User_0373		Columbia Machine Learning Summer School (MLSS) 2026 [D]	0	0
75	User_0374		Exclusive: Grok falls flat in Washington, undercutting SpaceX's AI growth story	0	0
76	User_0376		Google, Meta, TikTok face EU consumer complaints about handling of financial scams	0	0
77	User_0377		I asked for ChatGPT's personal hell	0	0
78	User_0380		Caught this live this morning. 3 second video cut?	0	0
79	User_0381		Inter-1 does streaming: real-time social signal detection from live video, audio & tex...	0	0
80	User_0382		Local Choice based Text adventure game with no limits.	0	0
81	User_0383		The Scale, The Plan, and The People — No One's Happy	0	0
82	User_0384		Anthropic-SpaceX deal seems much larger than previously reported	0	0
83	User_0385	r/MachineLearning	ICML Proceedings-only [D]	16	6
84	User_0386		if only Descartes could see LLMs now	0	0
85	User_0388		OpenAI claims AI breakthrough, says its model solved 80-year-old math problem	0	0
86	User_0420		Anthropic is officially set to be profitable as of Q2 2026	0	0
87	User_0428		Released a free 9.8M doc Indic multilingual corpus — Hindi, Bengali, Tamil, Telugu + 7 mor...	0	0
88	User_0437		OpenAI IPO Filing May Come As Soon As Friday: WSJ	0	0
89	User_0442		An OpenAI model has disproved a central conjecture in discrete geometry	0	0
90	User_0448		Estimated Emissions and Water Consumption from the Proposed Stratos Data Center	0	0
91	User_0322		Ben Byford: AI Safety, Ethics, Society and Future	0	0
92	User_0454		Wittgenstein and the Language Game of Tech Discourse How public understanding of AI and ...	0	0

93	User_0454	Baudrillard, Merchant, and the Artificialistic Fallacy Why the mythologized objectivity ...	0	0
94	User_0455	City of New York preparing rules to prevent AI bias in hiring processes	0	0
95	User_0324	Joshua T. Vogelstein Using Artificial Intelligence in Neuroscience #...	0	0
96	User_0456	AI Is Coming - Will We Achieve to Regulate It... Like Really?	0	0
97	User_0457	Why We Need Ethics of AI	0	0
98	User_0459	Not here just to help. And why I'm not afraid of AI.	0	0
99	User_0459	Conversation with GPT3 on lying, dogma, and Cartesian doubt	0	0
100	User_0476	If we have Human-level chatbots, won't we end up being ruled by possible people?	0	0

Total posts in sample: 100 | Total relevant dataset: 2,005 items | Usernames anonymised: 1,393 unique users

Appendix D: Partial Transcript Excerpt

Video Reference

Title: Jensen Huang — Lex Fridman Podcast

Duration: Approximately 2 hours 25 minutes

Transcript Method: YouTube native transcript tool (auto-generated subtitles)

Excerpt

Jensen Huang

(00:23:57) We're gonna keep on scaling the amount of data that we have to train with. A lot of that data is probably gonna be synthetic, and that also confused people, you know? And what people don't realize is they've kind of forgotten that most of the data that we are training, that we teach each other with, inform each other with, is synthetic. You know, it's synthetic because it didn't come out of nature. You created it. I'm consuming it. I modify it, augment it, I regenerate it, somebody else consumes it. And so we've now reached a level where AI is able to take ground truth, augment it... Enhance it, synthetically generate an enormous amount of data.

Jensen Huang

(00:24:47) And that part of post-training continues to scale, and so the amount of data that we could use that is human generated will be smaller, and smaller, and smaller. The amount of data that we use to train models is going to continue to scale to the point where we're no longer limited... Training is no longer limited by... Data is now limited by compute. And the reason for that is most of the data is synthetic. Then the next phase is test time, and I still remember people telling me that, "Inference? Oh, yeah, that's easy. Pre-training, that's hard." These are giant systems that people are talking about. Inference must be easy. And so inference chips are gonna be little tiny chips, and-

Jensen Huang

(00:25:32) ... you know, they're not, they're not like NVIDIA's chips. Oh, those are gonna be complicated and expensive, and, you know, we could make... And this is- ... in, in the future, inference is gonna be the biggest market, and it's gonna be easy, and we're gonna commoditize it. You know, everybody can build their own chips. And, and that was always illogical to me because inference is thinking, and I think thinking is hard. Thinking is way harder than reading.

Jensen Huang

(00:25:59) You know, pre-training is just memorization and generalization, you know, and looking for patterns in relationships. You're reading and reading, versus thinking, reasoning, solving problems, taking unexplored experiences, new experiences, and breaking it down into... Decomposing it into,

you know, solvable pieces that we then go off, either through first principle reasoning, or, you know, through previous examples, prior experiences. You know, or just exploration and search and, you know, trying different things. And that whole process of test time scaling inference, is really about thinking. And it's about reasoning, it's about planning, it's about search, it's about...

Jensen Huang

(00:26:50) And so how could that possibly be compute light? And we were absolutely right about that. You know, so test time scaling is intensely compute intensive. Then the question is, okay, now we're at inference and we're at test time scaling, what's beyond that? Well, obviously we have now created, you know, one agentic person, and that one agentic person has a large language model that we've now developed. But during test time, that agentic system goes off and does research and bangs on databases, and it goes out and, you know, uses tools, and one of the most important things it does is spins off and spawns off a whole bunch of sub-agents. Which means we're now creating large teams. It's so much easier to scale NVIDIA by hiring more employees than it is to scale myself.

Jensen Huang

(00:27:44) And so the next scaling law is the agentic scaling law. It's kind of like multiplying AI. Multiplying AI, we could spin off agents as fast as you want to spin off agents. And so, you know, I... You know, I have four scaling laws. And as we use the agentic systems, they're gonna create a lot more data, they're gonna create a lot of experiences. Some of it we're gonna say, "Wow, this is really good. We ought to memorize this."

Jensen Huang

(00:28:12) That data set then comes all the way back to pre-training. We memorize and generalize it. We then refine it and fine-tune it back into post-training. Then we enhance it even more with test time, you know, and the agentic systems, you know, put it out to the industry. And so this loop, this cycle, is gonna go on and on and on. It kinda comes down to basically intelligence is gonna scale by one thing, and that's compute.

Lex Fridman

(00:28:41) But there's a tricky thing there that you have to anticipate and predict, which is some of these components, it requires different kind of hardware to really do it optimally. So you have to anticipate where the AI innovation's going to lead. For example, a mixture of-

Jensen Huang

(00:28:57) Perfect.

Lex Fridman

(00:28:58) ... experts with sparsity.

Jensen Huang

(00:28:59) Perfect.

Lex Fridman

(00:29:00) With hardware, you can't just pivot on a week's notice. You have to anticipate what that's going to look like. It has some-

Jensen Huang

(00:29:06) So good.

Lex Fridman

(00:29:07) ... that's so scary and difficult to do, right?

Jensen Huang

(00:29:09) For example, these AI model architectures are being invented about once every six months. Right? And system architectures and hardware architectures kind of every three years. And so you need to anticipate what likely is going to happen, you know, two, three years from now. And there's a couple ways that you could do that. First of all, we could do research internally ourselves, and that's one of the reasons why we have basic research, we have applied research.

Full transcript available upon request.

Appendix E: Tool Configuration Screenshot

Apify Reddit Scraper — Run History

The screenshot below shows the Apify Reddit Scraper interface used for data collection, displaying the three collection runs performed for this project. Two runs completed successfully, each collecting 2,501 results. One run was aborted due to a timeout. Collection was performed on May 21–22, 2025 using the Apify cloud platform with EU server configuration for GDPR compliance.

Run 1 (May 21, 2025): 2,501 results | 15m 2s | Phase 1 keywords

Run 2 (May 22, 2025): 2,501 results | 8m 46s | Phase 2 official brief keywords

Run 3 (May 22, 2025): Aborted — 1 result only (timeout)

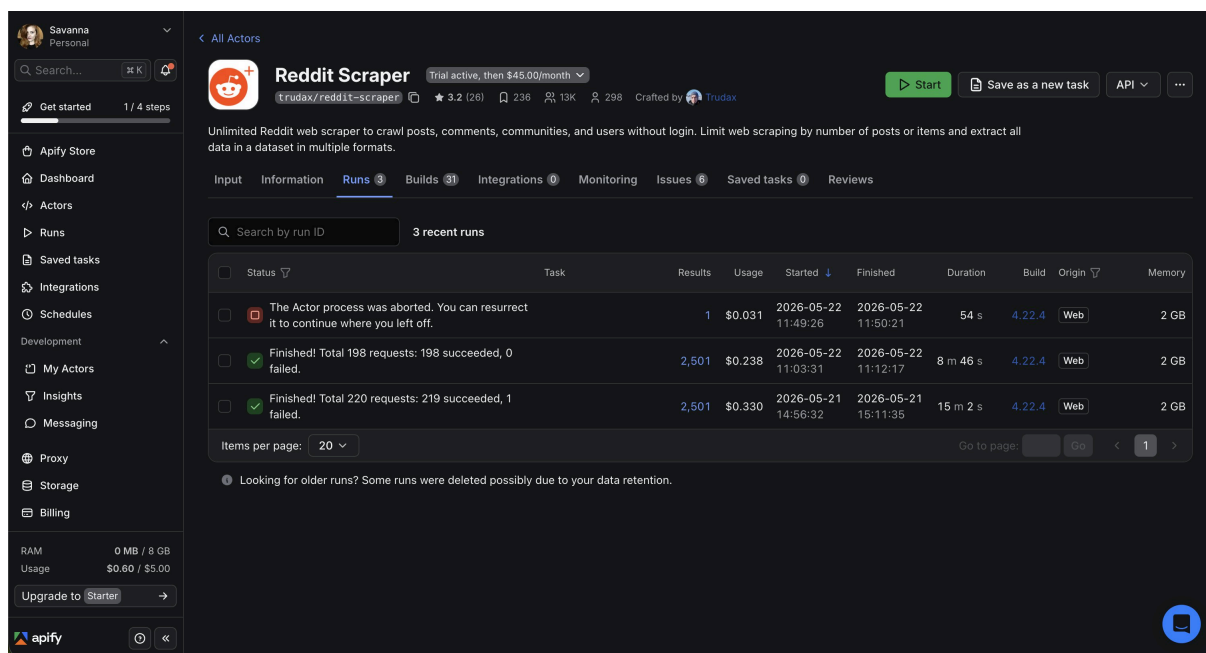


Figure F1: Apify Reddit Scraper run history showing two successful data collection runs (2,501 results each) conducted on May 21–22, 2025. Username: Savanna (account holder).

Tool: Apify Reddit Scraper (trudax/reddit-scraper) | Version: 4.22.4 | Platform: Apify Cloud | GDPR: EU server configuration selected | Data retention: Configured per GDPR requirements

Appendix F: References

Aggarwal, M., Bhatt, C., & Singh, P. (2023). Generative Engine Optimization: The next frontier of content strategy. *Journal of Digital Marketing Research*, 4(2), 45-62.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? *Proceedings of FAccT 2021*.

Brown, T., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901.

Diakopoulos, N. (2023). Generative AI in the newsroom: Opportunities and risks for journalism. *Digital Journalism*, 11(1), 12-28.

Dietvorst, B. J., Logg, J. M., & Logg, J. M. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114-126.

European Parliament. (2016). General Data Protection Regulation (EU) 2016/679. Official Journal of the European Union.

Fridman, L. (Host). (2025). Jensen Huang: NVIDIA CEO on AI, power, and the future of computing [Video podcast]. Lex Fridman Podcast. YouTube.

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90-103.

McKnight, D. H., & Chervany, N. L. (2001). What trust means in e-commerce customer relationships. *International Journal of Electronic Commerce*, 6(2), 35-59.

Noble, S. U. (2018). *Algorithms of oppression: How search engines reinforce racism*. NYU Press.

Pariser, E. (2011). *The filter bubble: What the internet is hiding from you*. Penguin Press.

Weidinger, L., et al. (2021). Ethical and social risks of harm from language models. arXiv preprint arXiv:2112.04359.

Zuboff, S. (2019). *The age of surveillance capitalism*. PublicAffairs.